

1.

假设我们已经知道了takes的属性数目，参考fig1，这里假设是6个属性，且students表与takes表的共同属性是ID，那么可以通过以下语句模拟左外连接，思路是很简单，首先找到students与takes的交集，然后使用union并上只属于students的部分，难点是怎样凑出形式：

```
select students.*, takes.course_id, takes.sec_id, takes.semester, takes.year,
takes.grade
from students,takes
where students.ID = takes.ID
union all
select *, null as course_id, null as sec_id, null as semester, null as year, null
as grade
from students
where students.ID not in (
select ID
from takes
)
```

以下语句模拟全连接，在左外连接的基础上并上只属于takes的部分即可：

```
select students.*, takes.course_id, takes.sec_id, takes.semester, takes.year,
takes.grade
from students,takes
where students.ID = takes.ID
union all
select *, null as course_id, null as sec_id, null as semester, null as year, null
as grade
from students
where students.ID not in (
select ID
from takes
)
union all
select takes.ID, null as name, null as dept_name, null as tot_cred,
takes.course_id, takes.sec_id, takes.semester, takes.year, takes.grade
from takes
where takes.ID not in (
select ID
from students
)
```

2.

因为"on delete cascaded"的存在，所以当父表中的它用于指定当父表中的某一行被删除时，所有在子表中依赖于该父行的外键行也应该被自动删除。因此当元组被删除时，所

有managerID = 被删除的employeeID的元组也会被删除。也意味着所有以被删除元组作为经理的员工数据也会被删除掉。

3.

创建视图语句如下（参考fig1）：

```
create view tot_credits(year,num_credits) as
select takes.year, sum(course.credits)
from takes,course
where takes.course_id = course.course_id
group by takes.year
```

4

原查询相当于找出没有导师的学生，可以将表进行外连接（这里进行左外连接），然后对i_ID是否为空进行判断即可：

```
select distinct ID
from students left outer join advisor on students.ID = advisor.s_id
where advisor.i_ID is null
```

5.

```
select course_id, sec_id, semester, year, section.building, section.room_number,
time_slot_id, capacity
from section inner join classroom using (building, room_number)
```

6.

不会。因为A本来就有select权限，其权限不受其他授权影响，B的授权不会影响A，不会影响图结构。